Software Test Plan (STP) - Student Attendance Management System (SAMS) System (Sample)

Project: Student Attendance Management System (SAMS)

Version: 1.0

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Date: 01-11-2025

Status: Sample

1. Introduction

Purpose: This document defines the test plan for the Student Attendance Management System (SAMS) System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing. Scope: Testing covers Student Attendance Management System (SAMS) features such as authentication, attendance marking, attendance viewing, attendance editing, and report generation. Hardware maintenance and core universitying system internals are excluded.

References: Student Attendance Management System (SAMS) SRS v1.0, Design Specifications v1.0, Definitions: Student Attendance Management System (SAMS) (Student Attendance Management System), password/biometric (Personal Identification Number), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix).

2. Test Items

- Authentication (biometric, password)
- Role-based access (student, faculty, admin)
- Attendance marking by faculty
- Attendance viewing by students
- Attendance editing with justification
- Reporting (downloadable reports, defaulter lists)

3. Features to be Tested

Features mapped to SRS requirement IDs:

Feature	SRS ID	Description	Acceptance Criteria
Account lockout after failed login attempts	SAMS-F-003	The system shall lock accounts after 5 failed login attempts.	After 5 failed attempts, login is blocked and a notification is sent.
Faculty attendance marking and editing	SAMS-F-010	Faculty shall be able to mark attendance in real-time via mobile and web apps.	Attendance saved instantly after marking.
	SAMS-F-012	Faculty shall be able to edit attendance with justification and audit logging.	Edits are saved only with mandatory justification and logged for auditing.
Real-time student attendance view	SAMS-F-011	Students shall be able to check attendance percentage and subject-wise breakdown.	Student view shows correct percentage per subject in real-time.
	SAMS-F-013	The system shall display daily presence/absence records to students, updated in real-time.	Day-wise daily status shown correctly.
Report generation and filtering	SAMS-F-020	The system shall allow students to view course-wise and overall attendance percentage.	Student logs in and sees their attendance percentage per course and overall.
	SAMS-F-021	The system shall allow faculty to download attendance reports in CSV/PDF formats.	Faculty selects course and date, system generates downloadable report file accurately.
	SAMS-F-024	The system shall allow faculty to filter attendance reports by course or date range.	Faculty applies filters and filtered reports are generated correctly.
Defaulter list generation based on thresholds	SAMS-F-022	The system shall allow administrators to generate defaulter lists for students below attendance thresholds (e.g., 75%).	Admin sets threshold and system outputs a list of low attendance students.

4. Features Not to be Tested

- Core university backend system logic (vendor responsibility)
- SAMS hardware device firmware
- External third-party integrations (payment gateways, etc.)

5. Test Approach / Strategy

Levels:- Unit tests (module-level)

Integration tests (Student Attendance Management System (SAMS)

-System tests (end-to-end Student Attendance Management System (SAMS) functionality)

Acceptance tests Types: - Functional testing (core features)

Usability testing (UI clarity, accessibility)

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password/biometric handling (masking, no logging)

- TLS 1.2+ verification- PCI-DSS compliance checks

- Fuzzing for input fields (login credentials data, password/biometric entry)

- Penetration testing of authentication flows

6. Test Environment

Hardware: Student Attendance Management System (SAMS) terminal with login credentials reader, dispenser, printer, attendance editing acceptor.

Software: Student Attendance Management System (SAMS) app v1.0, core university Ing API sandbox.

Tools: Streamlit (UI automation), Jira (defect tracking).

Test Data: login credentials, transactions.

7. Test Schedule

Milestones: - Test case design: Oct-2025

- Environment setup: Oct-2025

- Test execution start: Oct-2025

- Test execution end: 20-Nov-2025

8. Test Deliverables

- Test Plan (this document)

- Test Cases (manual & automated)

- Test Scripts

- Defect Reports

- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Chandana Lakya	Prepare plan, coordinate execution
Test Engineer	Chitra Madarakhandi	Design & execute test cases, log defects
Developer	Deesha C	Support defect fixes during testing
Product Owner	Daneshwari Melshetty	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Maintain backup environment on cloud VM
Dependency on third-party ATM hardware vendor	Engage vendor early and maintain test stubs

11. Assumptions & Dependencies

- Core university API will be stable and available

- Test data (accounts/login credentials) will be provided before execution

12. Suspension & Resumption Criteria

Suspend testing if: - Environment unavailable for >4 hours- Build is too unstable (blocks >30% test cases

) Resume testing if:- Blocking defects are resolved

- Environment stabilized

13. Test Case Management & Traceability

Requirement Description	SRS ID	Mapped Test Case ID	Test Case Description	Acceptance Criteria
Account lockout after failed login attempts	SAMS-F-003	TC-Auth-02	Validate handling of incorrect PIN entries and account lockout after 3 failed attempts	User account locks after 3 consecutive failed PIN entries; appropriate message shown
Faculty attendance marking and editing	SAMS-F-010, SAMS-F-012	TC-Att-01, TC-Att-02	Faculty marks attendance and edits with justification	Attendance recorded/edited successfully; logs updated with justification
Real-time student attendance view	SAMS-F-011, SAMS-F-013	TC-View-01, TC-View-02	Student views attendance data in real time	Attendance data displayed correctly and updated immediately upon change
Report generation and filtering	SAMS-F-020, SAMS-F-021	TC-Report-01	Generate attendance report, apply filters, and download in formats	Correct report data generated; filters applied accurately; downloadable files are correct
Defaulter list generation based on attendance thresholds	SAMS-F-022	TC-Defaulter-01	Generate defaulter list for students below threshold	List accurately reflects students with attendance below the set threshold

14. Test Metrics & Reporting

Metrics collected:- % test cases executed

- % passed/failed

- Defect density

- Defect aging

- Requirement coverageReports:- Daily execution status

- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		

16. Test Cases

This section provides sample test cases for the Student Attendance Management System (SAMS) System. Each test case includes a unique identifier, description, preconditions, input data, expected results, and postconditions.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Remarks
TC-Auth-01	Validate correct PIN entry	Valid student/faculty credentials	1. Enter valid username and PIN	Access to system granted	User session starts
TC-Auth-02	Incorrect PIN handling	Valid credentials	1. Enter wrong PIN 3 times	Account locked, error shown	Admin unlock required
TC-Att-01	Faculty marks attendance	Faculty logged in	1. Select class and students	Attendance recorded successfully	
TC-View-01	Student views attendance	Student logged in	1. Open attendance dashboard	Correct attendance details shown	
TC-Edit-01	Faculty edits attendance	Logged in as faculty	1. Modify attendance with justification	Edits saved and logged	
TC-Report-01	Generate attendance report	Admin logged in	1. Generate report with filters	Report generated accurately	
TC-Sec-01	Session timeout	User logged in, inactive 60s	1. No activity for 60s	Session logs out, re-login required	
TC-Perf-01	Response time check	Stable network, system online	1. Perform report generation	Response within 5 seconds	Meets SLA

Notes for Students:

• Maintain a consistent test case ID format (e.g., TC-Module-XX)• Each test case should trace back to a requirement ID from the SRS• Include both functional and non-functional test cases• Modify the sample table according to your project’s domain and functionality• Add columns such as 'Actual Result' and 'Status (Pass/Fail)' during execution